



A method of emotion contagion for crowd evacuation

Mengxiao Cao, Guijuan Zhang^{*}, Mengsi Wang, Dianjie Lu, Hong Liu

School of Information Science and Engineering, Shandong Normal University, Jinan 250014, China
Shandong Provincial Key Laboratory for Distributed Computer Software Novel Technology, Jinan 250358, China

HIGHLIGHTS

- An emotion contagion method for crowd evacuation is proposed.
- A P-SIS emotional contagion model is constructed.
- The P-SIS model is coupled with the social force model.
- A photo-realistic rendering method is used to exhibit crowd evacuation results.

ARTICLE INFO

Article history:

Received 27 December 2016

Received in revised form 6 March 2017

Available online 5 May 2017

Keywords:

Personality

P-SIS emotional contagion model

Social force model

Crowd evacuation

ABSTRACT

The current evacuation model does not consider the impact of emotion and personality on crowd evacuation. Thus, there is large difference between evacuation results and the real-life behavior of the crowd. In order to generate more realistic crowd evacuation results, we present a method of emotion contagion for crowd evacuation. First, we combine OCEAN (Openness, Extroversion, Agreeableness, Neuroticism, Conscientiousness) model and SIS (Susceptible Infected Susceptible) model to construct the P-SIS (Personalized SIS) emotional contagion model. The P-SIS model shows the diversity of individuals in crowd effectively. Second, we couple the P-SIS model with the social force model to simulate emotional contagion on crowd evacuation. Finally, the photo-realistic rendering method is employed to obtain the animation of crowd evacuation. Experimental results show that our method can simulate crowd evacuation realistically and has guiding significance for crowd evacuation in the emergency circumstances.

© 2017 Published by Elsevier B.V.

1. Introduction

Recently, emergencies in public spaces have occurred frequently which results in a considerable loss to life and property. Modeling and simulating crowd behaviors has been an active research topic in recent years. The crowd evacuation is affected by natural factors such as emotion. Conducting evacuation drill cannot realistically reflect the situations in emergency circumstances because people will not feel confused and panic during this process. Using computer simulation technology for simulating the crowd evacuation can not only reduce the costing, but also improve efficiency and accuracy. Therefore, adopting computer technology to simulate the crowd evacuation is of great significance.

Emotion is a kind of psychological activity produced by individuals along with cognition and consciousness, which not only affects individual's behavior, but also can affect the behaviors of other individuals through emotion contagion. Personality is a unique psychological tendency and individuals with different personalities have different possibility to be infected by emotions of other individuals. Therefore, simulating the crowd evacuation under the influence of emotion which

^{*} Correspondence to: School of Information Science and Engineering, Shandong Normal University, No.1 University Road, Jinan, 250358, China.
E-mail address: zhangguijuan@sdu.edu.cn (G. Zhang).

considers the personality of individuals is a very challenging issue. Although the SIS model [1] is classical and simple, it does not consider the influence of personality on emotional contagion.

To address the above problems, we present a novel crowd evacuation method based on emotion and personality. P-SIS emotion contagion model and social force model are two critical parts of this method. The personalized P-SIS model combines OCEAN model [2] and SIS model [1] to simulate the emotion contagion during crowd evacuation. The desired velocity of individual is modified so as to describe their movement under the influence of emotion. The experimental results show that this method can simulate the realistic crowd behavior under emotional contagion in the crowd evacuation.

Related work

1.1. Emotional contagion models

Researchers in VU University [3] proposed the ASCRIBE (Agent-based Social Contagion Regarding Intention Beliefs and Emotions) model. This model is a prototypical example of the heat dissipation phenomena studied in thermodynamics. Unfortunately, the ASCRIBE model is complex and does not consider the influence of personality. Bordas [4] presented an interactive installation to explore social emotional contagion. The model is used to simulate the propagation of emotions through artificial characters in an artificial environment. But the model ignores value analysis of emotional contagion. Margaux [5] put forward a model of emotional contagion based on individual personality and relationship in crisis situations, but the model just numerically describes emotional contagion, and does not perform in crowd simulation. Gratch [6] proposed the EMA model and defined a framework between individual emotion and social relationship. However, the model only supports small-scale movement. Dodds [7] presented the emotional contagion model based on epidemiology and divide people into two types: Susceptible Infected Recover (SIR) and Susceptible Infected Recover Susceptible (SIRS). These models have many emotional states which are inconvenient for calculation. Actually, in emergency situations, we only need to consider two emotional states: infected and susceptible. Therefore, we adopt SIS [1] model to describe the emotion contagion in crowd. We improve the model to consider the personality of individuals. This paper combines OCEAN model with SIS model and constructs the P-SIS personalized emotional contagion model for crowd evacuation.

1.2. Crowd simulation methods

Individuals have been attempting to diversify the movement behavior through psychological characteristics. For example, Refs. [8,9] simulated behavioral changes due to differences in personality. Liu Zhen [10] adopted the idea of intelligent individual to describe the individual behavior under the influence of emotion, and proposed a calculation method of emotional contagion. Kim et al. [11] put forward the crowd simulation method under the influence of dynamic psychology, which considered both the stable personality of individuals and the changes of crowd behavior when the environment changed. However, these models are mainly used to express the difference of individual movement in small groups rather than the large-scale crowd simulation.

Currently, there are two types of crowd evacuation model, macroscopic model and microscopic model. Macroscopic model takes global path planning and local collision conflict into account within a framework. Jiang et al. [12] used the global potential field model to simulate crowd evacuation. Wang et al. [13] proposed a crowd evacuation simulation method based on continuum crowd algorithm. Treuille et al. [14] presented a real-time crowd model based on the dynamic potential field. Narain et al. [15] introduced a novel constraint to model the large-scale behavior of the crowd, and accelerated inter-agent collision avoidance in dense scenarios. However, these methods have several drawbacks. First, they require high computational cost and large time consumption. Second, they do not consider the influence of emotion during crowd evacuation.

The microscopic model studies the crowd evacuation from the view of individual behavior, and balances its deficiencies in global search. At present, the most widely used crowd evacuation model is social force model [16], which can compute velocity and position of each individual and reproduce the phenomenon of “faster-is-slower” and “arching at the exit” in the process of evacuation. Many scholars have modified social force model for applying it to complex scenarios. Han et al. [17] introduced the information transmission mechanism into the social force model, and simulated pedestrian behavior in an emergency, especially when most pedestrians are unfamiliar with the evacuation environment. Ji et al. [18] combined the modified social force (MSFM) model and cellular automata (CA) model. Kretz et al. [19] investigated under which circumstances, parameter choices, and model variants oscillations do occur and how this can be prevented. Johansson et al. [20] introduced the social force model to produce waiting behavior and presented a sensitivity analysis and provide sufficient criteria for stability. Yang et al. [21] added leaders to the crowd, and produced the attractive force that drove individuals to move toward the position of leaders. Sui et al. [22] added the cohesion among pedestrians and the nervousness factor based on the original social force, and simulated the evacuation process of passengers under emergency in subway station. Huang et al. [23] introduced the crowd’s psychology repulsive force and direction changing force on the basis of the original social force model, and improved the model of crowd walking behavior for normal situation. Priscila Saboia et al. [24] evenly partitioned an area formed by an annulus around individuals into grids, and synthetically considered various factors, such as the angles between grids and targets, whether obstacles occupying grids and so on. Also, they calculated the weight of grids and determined the direction of motion at the next step. However, these corrections do not consider personality and emotion.

Table 1
The parameters of P-SIS emotional contagion model.

Sign	Significance
E	Emotional value of individuals
d	Panic dose
D	Accumulate of panic dose
T	Threshold of accumulated panic dose
β	Emotional decay factor
R	Distance threshold for neighbors
S	Distance threshold for administrator
t	Time step
N	Number of individuals in the crowd
ET	Emotion threshold

This paper introduces emotion to the original social force model, and the velocity of individual is modified to adjust person's desire. In addition, it can modify the desired force in the original social force and simulate emotional contagion in crowd evacuation.

2. Personalized emotion contagion in the crowd

There will produce emotional contagion when the pedestrians gathered, and the personality affects the emotion contagion process. This paper combines the OCEAN personality model [2] with the SIS model [1] and comes up with the P-SIS model for simulating the emotional contagion during crowd evacuation.

2.1. The OCEAN personality model

Considerable controversy exists in personality research about how many personality traits there are. However, one popular model is the OCEAN model [2]. The model has five factors: openness, extroversion, agreeableness, neuroticism and conscientiousness. Hatfield [25] argues that individuals who are sensitive to their own emotional changes and who have higher emotional responses are more likely to be infected. Therefore, this paper only considers the factor of "Neuroticism". Actually, most individuals generally have two emotional states: infected and susceptible during crowd evacuation. We select SIS model here. In this paper, the intensity of the neuroticism is represented by $P_N \in [0, 1]$. We incorporate P_N into the SIS model to construct P-SIS emotional contagion model for calculating emotional values.

2.2. P-SIS emotional contagion model

This paper constructs P-SIS emotional contagion model, where S represents susceptible state and I represents infected state. If the individual is infected, it generates a random dose. When the individual in state S comes into contact with an individual in state I , there is some probability to enter I state. The model defines a variable to represent the emotional state of individuals, and individuals with different sensitivities have different emotional values. At the same time, the model defines the administrator whose emotion keeps unchanged. The administrator causes the emotion level of crowds with an exponential decay at a rate β , such that individuals may become susceptible again.

In the P-SIS emotional contagion model, several variables are defined for each individual, as shown in Table 1.

In particular, the parameters are defined as

$$d_i = \log -N(\mu_d, \sigma_d^2), \quad (1)$$

$$T_i = \log -N(\mu_T, \sigma_T^2), \quad (2)$$

$$D(i, R) = \sum_{j \in \text{Neighbor}(i)} d_j, \quad (3)$$

$$E_i(0) = P_N(i) E_i(0), \quad (4)$$

$$E(t) = E(t-1) - \beta E(t-1). \quad (5)$$

Here, i and j represent the individual; μ_d is the mean of variable d ; σ_d^2 is the variance of variable d ; μ_T is the mean of variable T ; σ_T^2 is the variance of variable T ; d and T are calculated by a pre-determined log-normal distribution function $\log -N$. $D(i, R)$ is the accumulated panic dose of individual i from its neighbors $\text{Neighbor}(i)$. Here, the neighbor is defined that individuals who are mutually visible and locate within distance R . When the distance between the administrator and an infected individual is less than S , the emotion of the infected individual will decay accordingly. $E(t)$ is the emotion value of infected individuals at time t . $E_i(0)$ represents the emotion of individual i at time $t = 0$. In particular, if i is an infected individual, $E_i(0) = 1$. Otherwise, $E_i(0) = 0$. When the emotion of an infected individual decays to a threshold ET according to Eq. (5), it become a susceptible one.

Table 2

The emotional value and emotional state.

Value of E	Emotional state
0	Clam
(0, 0.4]	Anxiety
(0.4, 0.8]	Panic
(0.8, 1.0]	Hysteria

The emotion value E is in the interval (0, 1] if an individual is infected and 0 otherwise. According to literature [26], the emotion value of infected individuals can be divided into four levels corresponding to four different emotional states. Individuals of different personalities may take different level of emotional states. The details are shown in Table 2.

2.3. Algorithm of P-SIS emotional contagion model

Algorithm 1. Personalized emotional contagion

Input: $\mu_q, \sigma_q^2, \mu_T, \sigma_T^2, R, S$

Output: $E_i(t), i = 1, \dots, N$

Step 1 Initialize the sensitivity of all individuals with $P_N(i) = \text{rand}[0, 1], i = 1, \dots, N$;

Step 2 Randomly divide all individuals into two categories: susceptible or infected;

Step 3 Initialize all susceptible individuals: $E(0) = 0, d = 0$. Calculate T with μ_T and σ_T^2 according to Eq. (2);

Step 4 Initialize all infected individuals: $E(0) = 1$. Calculate d with μ_d and σ_d^2 according to Eq. (1). Calculate T with μ_T and σ_T^2 according to Eq. (2);

Step 5 At each time step t , update the emotional value E of all individuals by performing the following operations:

(1) Update the emotional value of each susceptible individual i

Step 5.1 Determine the neighbors $\text{Neighbor}(i)$;

Step 5.2 Update $D(i, R)$ according to Eq. (3);

Step 5.3 If $D(i, R) \geq T_i$, the individual i is infected. Calculate the emotion according to Eq. (4);

(2) Update the emotional value of each infected individual i

Step 5.4 Calculate the distance a_{dis} between the individual and the administrator.

Step 5.5 If $a_{\text{dis}} \leq S$, then the emotional value is updated by using decay factor β according to the Eq. (5);

Step 5.6 If $E_i < ET$, then the individual i becomes a susceptible one.

Algorithm 1 shows the emotion contagion process in a crowd. The infected individual generates a random dose to susceptible neighbors. If the cumulative doses of an individual exceeds the threshold T , the individual will be infected. However, the infected individual may also become susceptible under the influence of the administrator.

3. The crowd evacuation model with personalized emotion

In this section, we incorporate the P-SIS model with the social force model, and simulate crowd motion with emotions. In social force model, emotion changes the velocity of individual by the forces, further to alter crowd evacuation results. According to Newton's second law of motion, the acceleration of each individual i can be described as

$$m_i \frac{d\vec{v}_i(t)}{dt} = \vec{f}_i^0 + \sum_{j(\neq i)} \vec{f}_{ij} + \sum_w \vec{f}_{iw}, \quad (6)$$

where

$$\vec{f}_i^0 = m_i \frac{v_i(t) \vec{e}_i^0 - \vec{v}_i(t)}{\tau_i} \quad (7)$$

i represents the individual, and m_i is the mass of individual i . $\frac{d\vec{v}_i(t)}{dt}$ represents the acceleration generated by the individual in time t . $\vec{f}_{ij}$ is the interaction force between individual i and j . $\vec{f}_{iw}$ is the interaction between individual i and walls. $\vec{f}_i^0$ is the goal-driven force, which represents the individual force for moving toward the target. v_i^0 is the desired speed of individual. $\vec{e}_i^0$ is the motion direction, and τ_i is “relaxation time” in order to adapt to actual walking speeds.

In this paper, the desired speed $v_i^{\text{new}}(t)$ and movement trend $\vec{e}_i^{\text{new}}$ in social force model are improved as shown in Eqs. (8) and (9).

$$v_i^{\text{new}}(t) = (1 - E_i(t))v_i^0(t) + E_i(t)v_i^{\text{max}}, \quad (8)$$

$$\vec{e}_i^{\text{new}} = \lambda \vec{e}_j^0 + (1 - \lambda) \vec{n}_{ij}, \quad (9)$$

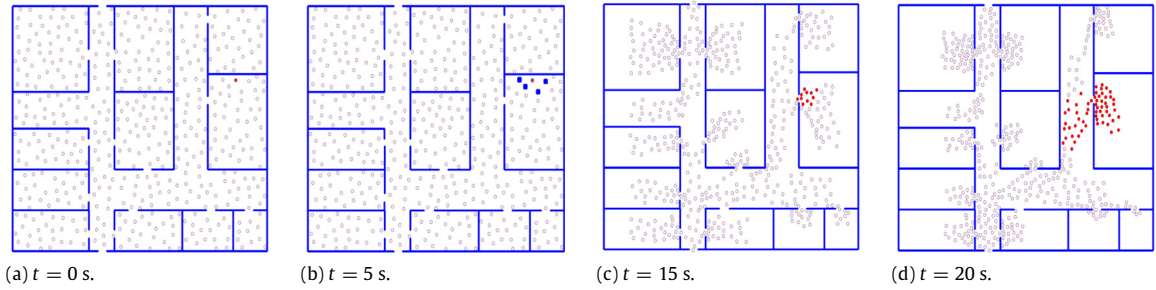


Fig. 1. Emotional contagion at different time in the office. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where $E_i(t)$ represents the emotion of individual i at time t , $v_i^{\max}$ is the maximum speed of individual i . λ is the weight, j is the individual that infects individual i . $\vec{e}_j^0$ represents the desired motion direction of the infected person j , $\vec{n}_{ij}$ is a vector pointing from individual i to j . The details of simulating crowd motion with emotions are shown in Algorithm 2.

Algorithm 2. Computing the velocity of individuals under emotional contagion model

Input: v_i^0 , $v_i^{\max}$, E_i , $\vec{e}_j^0$, λ

Output: $\vec{v}_i(t)$

Step 1 Initialize the crowd. Divide individuals into four types: clam, anxiety, panic and hysteria.

Step 2 At each time step t , update the velocity $\vec{v}_i$ of each individual i , do the following operations:

Step 2.1 Update the desired speed $v_i^{\text{new}}(t)$ according to Eq. (8);

Step 2.2 Compute the individual movement trend $\vec{e}_i^{\text{new}}$ according to Eq. (9);

Step 2.3 Compute the goal-driven force using v_i^{new} and $\vec{e}_i^{\text{new}}$ according to Eq. (7);

Step 2.4 Update the velocity $\vec{v}_i$ of the individual according to Eq. (6);

Step 3 If the individual fails to reach the goal position, it will continue to update the emotional value and velocity. Otherwise, the individual will stay near the goal position

After computing the velocity of each individual, we update its position with $x_i = x_i + \vec{v}_i \Delta t$. Algorithm 2 describes the velocity computation for individuals who are not administrators. The administrators move slowly than the other individuals because they are required to calm down the panic emotion of the crowd. In this paper, we set the velocity of the administrator to zero initially. When all the individuals leave the room, the administrator begins to move with the average human walking speed 1.4 m/s.

4. Results

We build a crowd evacuation system on a PC with 2.6 GHz Pentium dual-core processor and 4 GB memory. The experimental results are composed of two parts: the first part is the effect of P-SIS model and its influence on crowd evacuation; the second part is crowd evacuation and rendering effect on the platform of crowd evacuation [27]. In this experiment, the radius $r = 0.25$ m, the mass $m = 80$ kg, the desired velocity $v_i^0 = 0.8$ m s⁻¹, the constant in social force model $A = 2000$ N, $B = 0.08$ m, the large constant $K = 2.4 \times 10^5$ kg m⁻¹ s⁻¹, $k = 1.2 \times 10^5$ kg s⁻², $\alpha = 0.7$.

4.1. Emotional contagion

In order to analyze the influence of P-SIS model in evacuation, we simulate the emotional contagion and emotional decay. In this experiment, we set $\mu_d = 0.5$, $\sigma_d^2 = 2.0$, $\mu_T = 2.0$, $\sigma_T^2 = 7.0$.

① Emotional contagion in an office We do the evacuation experiment with 500 individuals in the office. Fig. 1(a) shows the initial condition. One can observe that an individual is infected (red color), whereas all the other individuals are susceptible (white color). In Fig. 1(b), the neighbors of an infected individual are marked in blue. The neighbors are individual who are mutually visible and locate with in a distance. So when the individuals are not in the same room, they are not neighbor even they the distance is small. In Fig. 1(c) and (d), the number of infected individuals increases during the evacuation process. When the infected individuals leave the room, the susceptible individuals (white color) in corridor may be infected.

② Emotional decay in the office We simulate crowd evacuation with 500 individuals in the office scene. Fig. 2 shows the results. Fig. 2(a) initializes the position of infected individuals and the administrator. We initialize infected individuals at a room and a corridor near the exit. An administrator (green color) is located at the room. Fig. 2(b) shows the evacuation state at $t = 5$ s. In Fig. 2(c), the emotion of the infected individual is decayed in the room when $t = 15$ s. We can see that

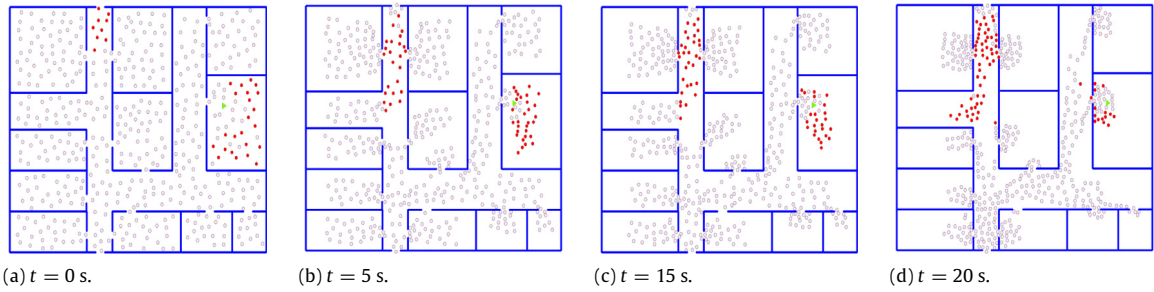
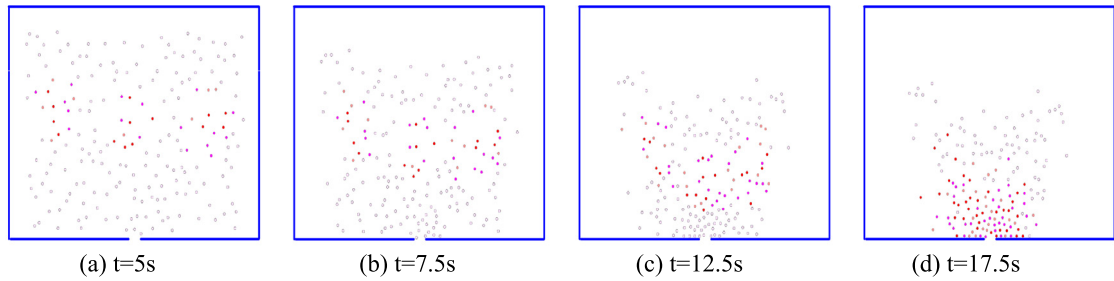
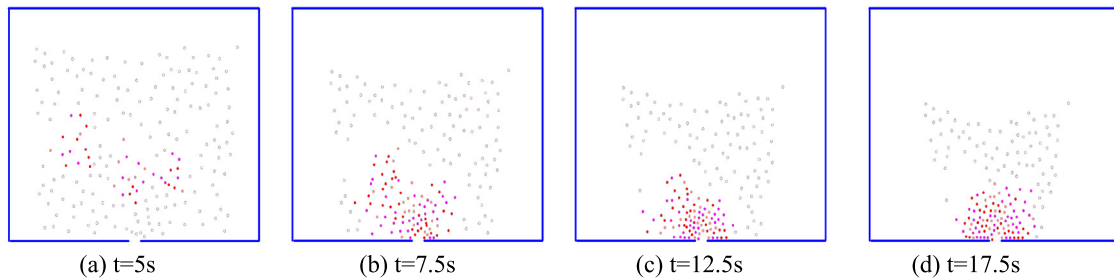


Fig. 2. Emotion decays at different time in the office. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(1) The positions of individuals without considering the influence of emotion at different time.



(2) The positions of individuals under the influence of P-SIS emotional contagion model at different time.

Fig. 3. The crowd simulation in a single-door room. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the individuals who are close to the administrator become susceptible again. In Fig. 2(d), some susceptible individuals are infected again because they are far away from the administrator.

In order to demonstrate the influence of the P-SIS model, this paper takes two experiments (③④). In these experiments, individuals are colored in white, orange, magenta, and red so as to describe different emotional state of individuals (i.e., calm, anxiety, panic and hysteria).

③ Crowd simulation in a single-door room We make comparisons between the motion of the crowd with or without considering emotions in the single-door room. Fig. 3 shows an example with 200 individuals in the single-door room. Fig. 3(2)(b) shows the infected individuals move to the exit more quickly than the susceptible individuals. In addition, individuals in hysteria state (red color) are in front of other infected individuals. In Fig. 3(2)(c), an arch is formed at the exit by infected individuals. All these individuals are walking in a normal speed to the exit in Fig. 3(1)(b) and (1)(c). In Fig. 3(2)(d), some infected individuals have left the room when $t = 17.5$ s, whereas these individuals are moving toward the exit in Fig. 3(1)(d). The experiment shows that emotional contagion model can simulate real-world situation more realistically if some individuals are panic in the crowd evacuation process.

④ Crowd simulation in the office We conduct a comparison experiment with 500 individuals in the office shown in Fig. 4. When $t = 20$ s, some infected individuals have left the room in Fig. 4(2)(a) and these individuals are moving to the exit of room in Fig. 4(1)(a). In Fig. 4(2)(b) and (2)(c), more individuals are infected during evacuation process compared to Fig. 4(1)(b) and (1)(c). In Fig. 4(2)(d), the individuals remaining in the office is fewer than those of Fig. 4(1)(d), because the individuals with panic emotion runs more faster than the normal speed. The experiments that the rationality of dividing the emotion level and the P-SIS emotional model simulates crowd evacuation realistically.

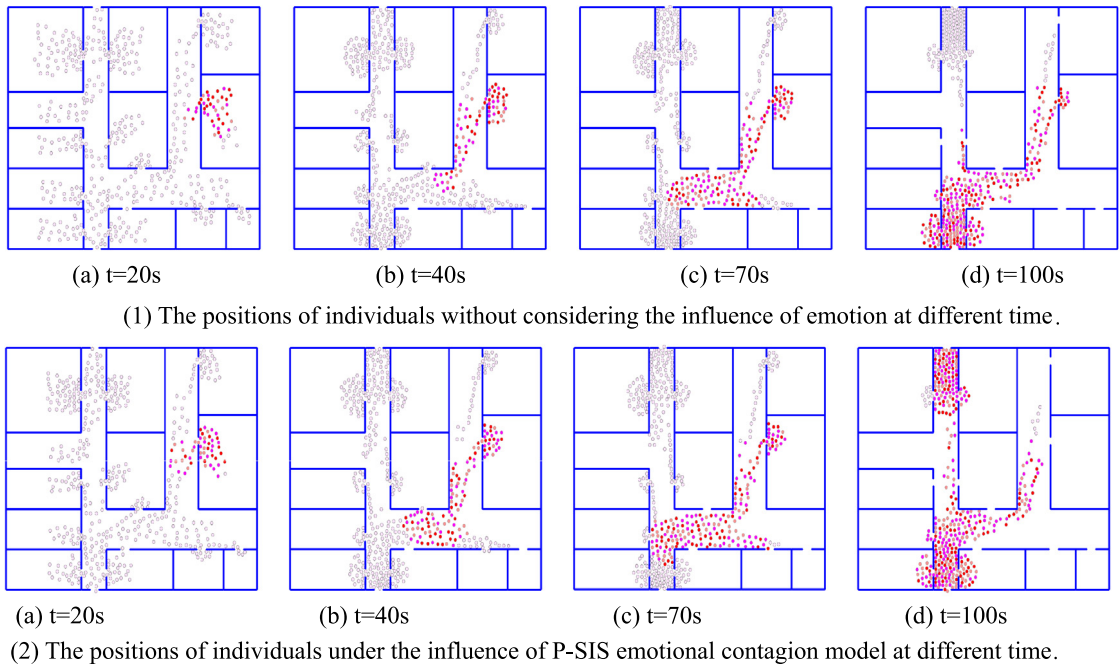


Fig. 4. The crowd simulation in the office.

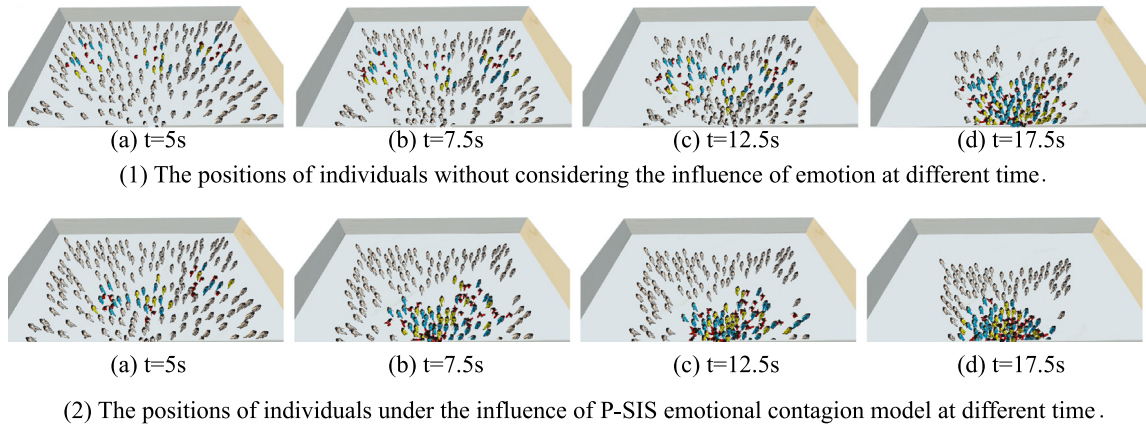


Fig. 5. Crowd simulation in the single-door room. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Crowd evacuation with emotional contagion

We introduce XNA framework [18] and use XNA Game Studio 2013 to generate the crowd animation of scenario ③ and ④ respectively, where individuals in clam (white color), anxiety (blue color), panic (yellow color) and hysteria (red color) state.

Figs. 5 and 6 obtain the crowd animation of method ③ and ④ in Section 4.1. Fig. 5 shows that the experimental results are in consistent with Fig. 3, and Fig. 6 shows that the experimental results are in consistent with Fig. 4. The experiment proves the influences of emotional contagion model on the crowd evacuation. It also demonstrates that it is reasonable to divide the emotion into different levels. The experiment shows that crowd simulation based on emotional contagion model can produce more realistic results.

We show the quantitative analysis about the influence of emotion contagion model in Figs. 7 and 8. In this experiment, the population size varies from 50 to 600. The average evacuation time is shown in Fig. 7. It indicates that the evacuation time for crowd with emotion is less than the crowd without emotion, because infected individuals move faster than susceptible individuals. When the population size is small, the infected individuals form arch quick at the exit while the individuals without emotions does not form the arch at the exit. When the number of individuals increases, the infected individuals

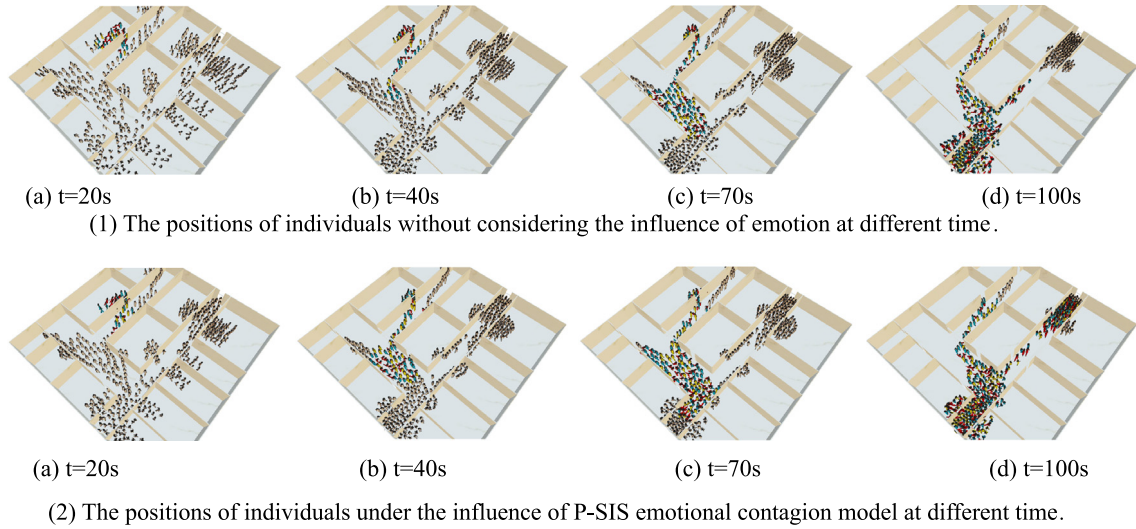


Fig. 6. Crowd simulation in the office. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

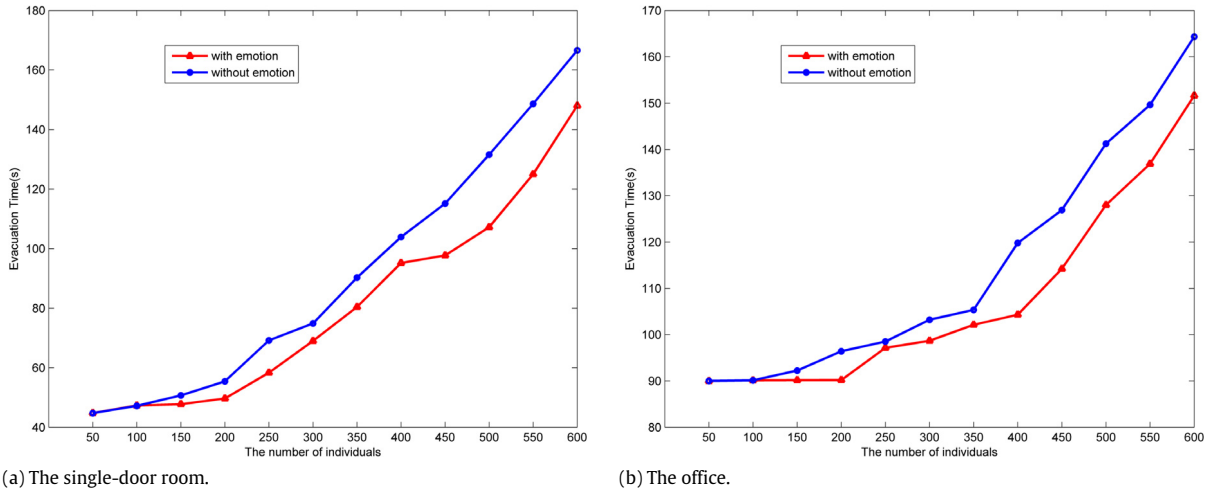


Fig. 7. The evacuation time in different scenarios.

and susceptible individuals both form the arch. However, the infected individuals move faster, and thus, the evacuation time is shorter. When the population size increases, the evacuation time increases accordingly. As we can see in above graphs, the emotion contagion model simulates the crowd evacuation realistically.

We also compute the efficiency of crowd evacuation. Here, efficiency is the number of evacuated individuals per second [28]. We compare the efficiency of crowd evacuation with and without considering the influence of emotion in Fig. 8(a). It demonstrates that the efficiency of evacuating crowd with emotion is higher than the efficiency of evacuating crowd without emotion. The evacuation efficiency increases with the size of population. It reaches a maximum when population size is 450 for crowd with emotion. But the maximum efficiency is achieved when population size is 300 if the emotion is not considered. We also compare the evacuation efficiency in a same room but set different number of doors. See Fig. 8(b), the more exits, the higher the efficiency of evacuation. The efficiency of crowd evacuation reaches maximum at different size of population. For example, the population size is 450 for one door scenario, and 500 for two doors scenario. However, the evacuation efficiency in the four-door scenario is still not reached the peak when population size is 600. Therefore, the scenario with four doors is more suitable for large-scale crowd evacuation.

5. Conclusions

Aiming at improving the original social force model which ignores the personality and emotion during crowd evacuation, we put forward an evacuation model based on emotional contagion. This paper introduces emotion to the original social

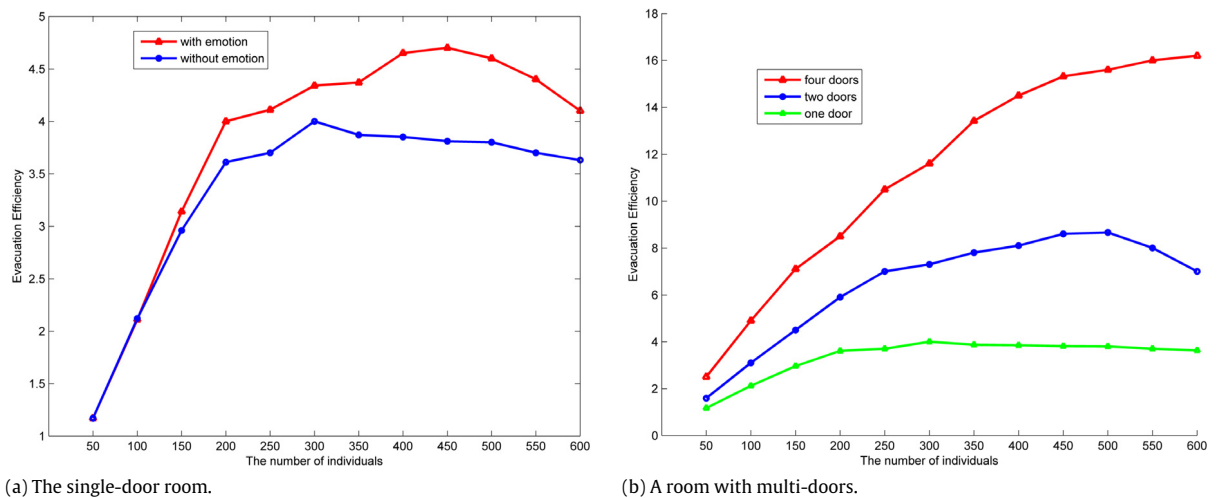


Fig. 8. The comparison of crowd evacuation efficiency in different scene.

force model, and the desired velocities of individuals are modified for describing their motions under different emotion. This phenomenon simulates the real scenario where the emotion may spread in the crowd. Therefore, this research not only develops a realistic modeling of pedestrian flow but also is important for a better preparation of emergency management. We would like to extend our work to consider social relationships in our future work.

Acknowledgments

This work is supported by National Natural Science Foundation of China under Grant Nos. 61572299, 61402270, 61472232, 61373149, 61402269, and Shandong Provincial Natural Science Foundation of China under Grant No. ZR2014FQ009.

References

- [1] P.S. Dodds, D.J. Watts, A generalized model of social and biological contagion, *J. Theoret. Biol.* 232 (4) (2005) 587–604.
- [2] R.R. McCrae, P.T. Costa, Personality in adulthood: a five-factor theory perspective, *Int. Cultural Psychol.* (2002) 303–322.
- [3] J. Tsai, E. Bowring, S. Marsella, et al., Empirical evaluation of computational fear contagion model crowd dispersions, *Auton. Agents Multi-Agent Syst.* 27 (2) (2013) 200–217.
- [4] Jeremie. Bordes, OMNEMOTION: the propagation of emotions// VRIC (Virtual Reality International Conference). 2014, PP. 1–4.
- [5] M. Lhommet, D. Lourdeaux, J.P. Barthés, Never alone in the crowd: A microscopic crowd model based on emotional contagion. in: IEEE/WIC/ACM International Conference on Intelligent Agent Technology, lat 2011, Campus Scientifique De La Doua, Lyon, France, August Vol. 2, 2011, pp. 89–92.
- [6] J. Gratch, S. Marsella, A domain-independent framework for modeling emotion, *Cogn. Syst. Res.* 5 (4) (2004) 269–306.
- [7] P.S. Dodds, D.J. Watts, Universal behavior in a generalized model of contagion, *Phys. Rev. Lett.* 92 (21) (2004) 5697–5711.
- [8] F. Durupinar, N. Pelechano, J.M. Allbeck, et al., How the ocean personality model affects the perception of crowds, *IEEE Comput. Graph. Appl.* 3 (2011) 22–31.
- [9] S.J. Guy, S. Kim, M.C. Lin, et al., Simulating heterogeneous crowd behaviors using personality trait theory, in: Proceedings of the 2011 ACM SIGGRAPH Eurographics Symposium on Computer Animation, ACM, 2011, pp. 43–52.
- [10] Liu Zhen, Jin Wei, Huang Peng, et al., An emotion contagion simulation model for crowd events, *J. Comput. Res. Dev.* 50 (12) (2013) 2578–2589.
- [11] M. Kim, K. Hyun, J. Kim, et al., Synchronized multi-character motion editing, *ACM Trans. Graph.* 28 (3) (2009) 79.
- [12] H. Jiang, W. Xu, T. Mao, et al., Continuum crowd simulation in complex environments, *Comput. Graph.* 34 (5) (2010) 537–544.
- [13] Wang Zhao-Qi, Mao Tian Lu, Jiang Hao, et al., Guarder: virtual drilling system for crowd evacuation under emergency scheme, *J. Comput. Res. Dev.* 47 (6) (2010) 969–978.
- [14] A. Treuille, S. Cooper, Z. Popovic, Continuum crowds, *ACM Trans. Graph.* 25 (3) (2006) 1160–1168.
- [15] R. Narain, A. Golas, S. Curtis, et al., Aggregate dynamics for dense crowd simulation, *ACM Trans. Graph.* 28 (5) (2009) 122. ACM.
- [16] D. Helbing, L. Buzna, A. Johansson, et al., Self-organized pedestrian crowd dynamics: Experiments, simulations, and design solutions, *Transp. Sci.* 39 (1) (2005) 1–24.
- [17] Y. Han, H. Liu, Modified social force model based on information transmission toward crowd evacuation simulation, *Physica A* (2016).
- [18] X. Ji, J. Zhang, Y. Hu, et al., Pedestrian movement analysis in transfer station corridor: Velocity-based and acceleration-based, *Physica A* 450 (2016) 416–434.
- [19] T. Kretz, On oscillations in the Social Force Model, *Physica A* 438 (2015) 272–285.
- [20] F. Johansson, A. Peterson, A. Tapani, Waiting pedestrians in the social force model, *Physica A* 419 (2015) 95–107.
- [21] X. Yang, H. Dong, Q. Wang, et al., Guided crowd dynamics via modified social force model, *Physical A* 411 (10) (2014) 63–73.
- [22] Sui Jie, Wang Jia-hui, Yu Hua, Research on simulation of emergency evacuation based on social force model, *J. Syst. Simul.* 26 (6) (2014) 1197–1201.
- [23] Huang Peng, Liu Zhen, Study on improved social force model for crowd simulation, *J. Syst. Simul.* 24 (9) (2012) 1916–1919.
- [24] Priscila Saboia, S. Goldenstein, Crowd simulation: applying mobile grids to the social force model, *Vis. Comput.* 28 (10) (2012) 1039–1048.
- [25] E. Hatfield, J.T. Cacioppo, R.L. Rapson, Emotional contagion: Studies in emotion and social interaction (1993).
- [26] Yu Li-Peng, Liu Zhen, L.I.U. Ting-Ting, et al., Sampling speed based emotion contagion model for panic crowd, *Appl. Res. Comput.* 32 (10) (2015) 3172–3178.
- [27] Sun Yun-Chen, Liu Hong, Zhang Gui-Juan, Cross platform system for real-time crowd simulation, *Chinese Comput. Syst.* 36 (4) (2015) 863–867.
- [28] Zhu Kong-Jin, Yang Li-Zhong, The effects of exit position and internal layout of classroom on evacuation efficiency, *Acta Phys. Sinica* 59 (11) (2010) 7701–7707.